

SALMAN SADIQ

UNITY DEVELOPER | VR, AR, ANDROID & MULTIPLAYER SYSTEMS

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PROFILE

Unity Developer and Associate Software Engineer with 2 years building VR, AR, Android, multiplayer, and AI-driven applications. Core stack: Unity/C#, Meta Quest 2/3, XR Toolkit, Unity Gaming Services, WebRTC, Azure TTS, Firebase, and .NET backends on Google Cloud. Recently shipped AI teachers with real-time lip-sync, a multiplayer VR meeting room for up to 10 Quest users, and **Empire Avenue**, an online Android board game released on itch.io with AdMob, Google Sign-In, and a self-hosted .NET game server.

EXPERIENCE

Associate Software Engineer

Ilmiversity

October 2025 - September 2026

Lahore, Pakistan

- Built conversational AI avatars with real-time lip-sync – three subject teachers for physics, chemistry and programming, plus a guide robot on each of six expo stalls – mapping Azure TTS visemes to facial blendshapes through a JSON-driven framework.
- Resolved the AI, TTS and STT provider and its credentials at runtime from Firebase Remote Config rather than baking them into the build, so providers can be switched and keys rotated without shipping a new APK.
- Sustained the target 72/90 FPS across every Meta Quest 3 scene using LODs, static/dynamic batching, occlusion culling, baked lightmaps, and shader-driven effects.
- Implemented in-world scene transitions that carry a visitor from a stall's miniature model into that model's full-scale environment to explore it.
- Engineered a multiplayer VR meeting room for up to 10 concurrent Quest users on Unity Gaming Services (Lobby and Relay) and Netcode for GameObjects.
- Integrated Unity Render Streaming with WebRTC to stream a presenter's browser screen into VR, building the Node signaling server, STUN/TURN configuration, and Firebase-backed client log collection that pinpointed why cross-network viewers failed to connect.
- Added Firestore-backed leaderboards across the physics and chemistry level flows, ranking student scores between sessions.
- Integrated Model Context Protocol (MCP) tooling into the Unity Editor for automated scene inspection and asset edits, and implemented Data Version Control (DVC) to version AI datasets and model artifacts.
- Delivered across two-week Jira sprints, reviewing every merge and tracking cross-team dependencies in Asana.

Games & VR Developer

UET Game Studio

April 2024 – September 2025

Lahore, Pakistan

- Built controller interactions and realistic steering wheel physics using Realistic Car Controller (RCC) and XR Toolkit.
- Developed wave spawning, state machines, and ragdoll systems for a multiplayer FPS on Opsive Character Controller, networked over Photon PUN.
- Shipped Unity titles across VR, mobile, PC, and WebGL, deploying browser builds to Vercel for instant cross-platform access.
- Streamlined team collaboration by rolling out Unity Version Control / Plastic SCM branching, cutting merge conflicts on shared scenes.

SKILLS

Programming	C#, OOP, C++, DSA, SQL
Game Development	Unity, Gameplay Programming, AI Systems, UI Systems, Animation, Timeline, Cinematics, Particle Effects
AR/VR	XR Toolkit, MRTK, Meta Quest 2/3, AR Foundation, Vuforia, VR Interaction Systems
Multiplayer & Web	Unity Gaming Services (Lobby, Relay), Netcode for GameObjects, Photon PUN, Unity Render Streaming, WebRTC, WebGL, Vercel
Backend & Cloud	Firebase (Remote Config, Firestore), .NET / ASP.NET Core, WebSockets, REST APIs, Server-Authoritative Multiplayer, Google Cloud Compute Engine, Linux (systemd), Caddy HTTPS
Android & Release	IL2CPP/ARM64 Builds, APK/AAB Signing, Addressables, AdMob Rewarded Ads (UMP Consent, Server-Side Verification), Google Sign-In, Play Policy Compliance, itch.io Publishing
AI & Speech	Azure TTS/STT, GPT & Gemini APIs, Conversational AI Chat, Viseme & Blendshape Pipelines
Optimization & Debugging	Unity Profiler, Frame Debugger, Build Report, ADB Logcat, LODs, Batching, Occlusion Culling, ASTC Compression, Baked Lighting, Mobile Shaders
Tools & Practices	Visual Studio, Git/GitHub, Plastic SCM, Jira, Asana, NUnit, Model Context Protocol (MCP), Data Version Control (DVC), Agile/Scrum, Code Reviews

PROJECTS

Empire Avenue – Online Android Board Game | *Personal Project* | *Unity 6* | *C#* | *.NET* | *Google Cloud* | *AdMob* | [Play on itch.io](#)

3D property-trading board game for 2–8 players with Pakistan, UK and US boards defined in JSON, released for Android on itch.io.

- Built a Unity-free C# rules engine shared by the game and a .NET dedicated server, covered by 450+ NUnit tests; online rooms with join codes, host approval, reconnect, and bots filling empty seats over WebSockets.
- Deployed the server to a Google Cloud VM (Linux, systemd, Caddy HTTPS, nightly backups), with the server address fetched at launch so builds never need rebuilding after a server move.
- Integrated guest accounts, Facebook Login (later migrated to Google Sign-In via Android Credential Manager), cloud progress sync, in-app account deletion, and AdMob rewarded ads with consent and server-side reward verification.
- Cut the APK from 527 MB to about 150 MB and removed scene-load freezes using the Profiler, Frame Debugger, Build Report and logcat: 202 materials moved to Simple Lit, ASTC compression, texture and mesh budgets, baked occlusion, and Addressables; verified 60 FPS on the board with zero janky frames.
- Audited the game against Google Play policy (trademarks, account deletion, target SDK 36, 16 KB page size) and versioned 800 MB of art with DVC.

Da1lImverse - VR Campus Platform | *Ilmiversity* | *Unity/C#* | *Meta Quest 3* | *Azure TTS* | *Firebase* | *UGS Lobby & Relay* | *WebRTC*

Enterprise VR campus delivered as one product across several environments: **V Campus**, an explorable campus tuned for standalone VR; **AI Classroom**, where three AI subject teachers lip-sync to their own speech and answer student questions through conversational chat, with the AI and speech providers selected through Firebase Remote Config; and a **Multiplayer Meeting Room** for up to 10 users joining by 6-digit code, with a presenter's browser screen streamed live into VR.

Da1Expo Hall | *Ilmiversity* | *Unity/C#* | *Meta Quest 3* | *Azure TTS* | *AI Chat*

VR expo with six themed stalls, each hosting a robot guide that answers questions about the miniature on display. A stall button carries the visitor into that model's full-scale environment – stepping from a resort miniature into the resort itself to walk through it.

Earlier Projects | *UET Game Studio*

- **Car VR Simulation** (*XR Toolkit, RCC*) – VR racing with hand-tracked steering physics, AI opponents, and 3 game modes. [Demo](#)
- **Waste Land of Living Dead** (*UFPS, Photon PUN*) – multiplayer zombie shooter with lobbies, wave spawning, ragdolls, and cinematics.
- **VR Bio Lab** (*MRTK*) – educational VR simulator for inspecting and manipulating anatomical models. [Demo](#)
- **ARPlace** (*AR Foundation, Android*) – AR furniture placement using touch gestures and plane detection. [Demo](#)
- **WebGL Titles** (*WebGL, Vercel*) – [Quiz the Globe](#), a flag-identification game with Time Trial and Survival modes, and [Letter Cascade](#), a word puzzle with physics-based letter collisions.

EDUCATION

Master of Computer Science | University of Okara

2021 – 2023

BSc Computer Science & Double Mathematics | University of the Punjab

2018 – 2020